

EduGenie – Google Gemini Powered Learning Assistant

Data Flow Diagram

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Level-1 Data Flow Description

Element	Type	Description
Student	External Entity	Provides question, topic, passage, quiz request, or learning-path request.
1. Validate Request	Process	Validates required fields and input structure.
2. Route Feature	Process	Routes the request to Q&A, explanation, summary, quiz, or learning-path module.
3. Gemini Service Layer	Process	Discovers available models, selects primary/fallback model, and sends the prompt.
Google Gemini API	External Entity / Service	Generates the requested natural-language response.
Configuration (.env)	Data Store	Holds configuration such as GEMINI_API_KEY outside source code.
Offline/Demo Explainer	Process	Provides simplified local output when live credentials/network are unavailable.
Student	External Entity	Receives structured result through the web frontend/API.

Flow

Student → Validate Request → Route Feature → Gemini Service Layer → Google Gemini API → Feature Module → Web Frontend → Student. If Gemini is unavailable, the service layer routes to the offline/demo fallback.